

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	Credit card Approval Prediction System

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Prediction accuracy depends on the quality and quantity of the training	Poor-quality or insufficient data may reduce prediction accuracy.	High
2	The system does not maintain user authentication or prediction history.	Users cannot securely access or review previous prediction records.	Medium
3	The application is deployed as a basic Flask web application without database integration.	Limited scalability and data management capabilities.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on scalable cloud infrastructure with load balancing
2	Data Storage	Uses a CSV dataset	Integrate a relational database such as MySQL or PostgreSQL
3	Performance	Uses a single trained machine learning model for prediction.	Optimize the model and implement caching for faster predictions
4	Security	Basic Flask application	Add user authentication, HTTPS, and input validation

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate and compare additional machine learning models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost.	2 Months	Improve prediction accuracy and model robustness through algorithm comparison and ensemble selection.
Phase 2	Expand the dataset with more recent and diverse applicant data and implement advanced feature engineering techniques.	4 Months	Enhance model performance and generalization by using high-quality and up-to-date data.
Phase 3	Add interactive dashboards, data visualizations, and analytics for insights on approval trends and applicant profiles.	6 Months	Better data visualization and insights for decision-making and business analysis.
Phase 4	Deploy a mobile-friendly application with user authentication, prediction history, and role-based access control.	9 Months	Improved accessibility, security, and scalability for end users and administrators.